

2(a)(i)	<i>any two from</i> 1 volume of urine ; 2 volume / concentration / composition / pH, of agar ; 3 concentration / volume, of antibiotics ; 4 (incubation) temperature ; 5 incubation time ; 6 size / diameter / area / spacing / type / source, of discs ;	2
2(a)(ii)	1 measure the, diameter / radius / area, of clear zone around the antibiotic disc (with a ruler / callipers / grid) ; 2 <i>idea of</i> the, larger / wider / bigger / AW ,the clear zone is, the, more sensitive / less resistant, the bacteria are to the given antibiotic / ora or idea of no clear zone means bacteria are, resistant to / not affected by / not killed by / not sensitive to, (given) antibiotic ;	2

Question	Answer	Marks
2(b)(i)	<i>any two from</i> 1 <i>idea that</i> it shows a large, spread of data (around the mean) / difference in data / deviation from the mean / variation with the mean ; 2 data is not (very) reliable / trustworthy / consistent / AW ; 3 standard deviation increases as mean increases / positive correlation between standard deviation and mean ;	2
2(b)(ii)	<i>any two from</i> <i>idea that</i> 1 patients may not take the antibiotics prescribed ; 2 patients may not complete the course of antibiotics ; 3 dose (per prescription) may differ ; 4 antibiotics are also prescribed by dentists / hospitals / other practitioners ; 5 patients may buy antibiotics (without a prescription) ; 6 data only taken from large clinics ; ora 7 patients may be prescribed, antibiotics other than those stated in table / more than one antibiotic ;	2
2(c)(i)	<i>any one from</i> 1 data, may be / is, non-linear / skewed ; 2 data, not / may not be, normally distributed ; 3 data is ordinal / discontinuous / not continuous / discrete ; 4 scatter, graph / diagram, shows / may show, that there is a relationship ; 5 data samples are independent ; 6 <i>ref. to</i> random selection ;	1

Question	Answer	Marks
2(c)(ii)	there is no (significant) correlation between antibiotic use / (number of) prescriptions and, the percentage of urine samples containing resistant <i>E. coli</i> / the presence of resistant strains of <i>E. coli</i> in the urine (of patients) ;	1
2(c)(iii)	1 two from ampicillin, trimethoprim, quinolone ; 2 calculated value / r_s, is higher than the, critical / table value (of r_s at, $p = 0.05$ / 0.362) or critical value is, at $p = 0.05$ / 5% significance / 0.362 ;	2
2(d)	<i>description max 2</i> percentage of (ampicillin-)resistance / (ampicillin-)resistant <i>E. coli</i> 1 is higher in males than females (of the same age / at all ages) ; ora 2 is initially high in, (very) young / 3-year olds / up to 15 ; 3 decreases (steeply), in young / up to (around) 20 ; 4 resistance, more or less plateaus / fluctuates / changes little / goes up and down, between ages 20 / 30 and 60 / 80 ; 5 increases steeply from (around) 80 or increases from around, 60–80 / old age ; 6 is highest at 100 years (in both) ; <i>explanation (must be linked to antibiotic use) max 2</i> 7 antibiotic / ampicillin, use is higher in the, young / old ; ora 8 idea of people between 20 and 60 / 80 are prescribed, different antibiotics / antibiotics other than ampicillin ; 9 males, may be less likely to complete the full course of antibiotics / may take more antibiotics ; ora 10 idea of young / old, have greater contact with, hospital / health care centres, and therefore greater exposure to, ampicillin / antibiotic-resistant <i>E. coli</i> ;	3